SIMATS ENGINEERING

ASSESSMENT TOOL 2 - Medium(Assignment 1)

COURSE CODE: CSA09 **COURSE NAME: Programming in Java COURSE OUTCOME COVERED**

CO2: *Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications.(BL3,BL4)*

Assessment Tool Weightage: 10%

QUESTIONS: 40 (each student assigned distinct questions) **Total Marks: 100**

ANNEXURE A

Assignment 1

Code of Conduct:

*I **N.Adi Narayana** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding				
Algorithm				
Code Implementation				
Competitive Performance				
Test Case Handling				

Annexure A

Assignment 1

Question 36 – Cyber Security Log

A company stores security event logs.

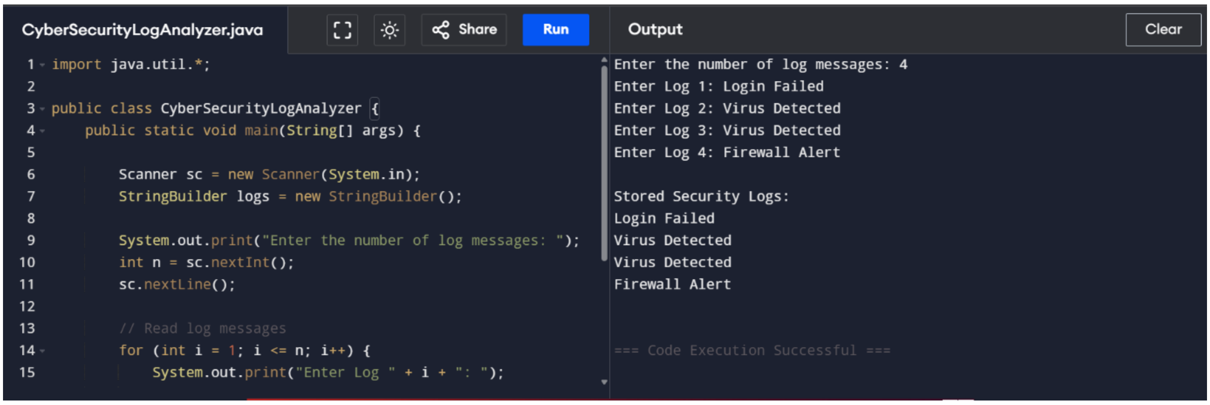
Questions

a) **Develop a Java program using StringBuilder.**

```
import java.util.*;


public class CyberSecurityLogAnalyzer {
    public static void main(String[] args) {


        Scanner sc = new Scanner(System.in);
        StringBuilder logs = new StringBuilder();


        System.out.print("Enter the number of log messages: ");
        int n = sc.nextInt();
        sc.nextLine();


        // Read log messages
        for (int i = 1; i <= n; i++) {
            System.out.print("Enter Log " + i + ": ");
            String log = sc.nextLine();
            logs.append(log).append("\n");
        }
```

```
        // Display all logs

        System.out.println("\nStored Security Logs:");

        System.out.println(logs);

    }

}
```

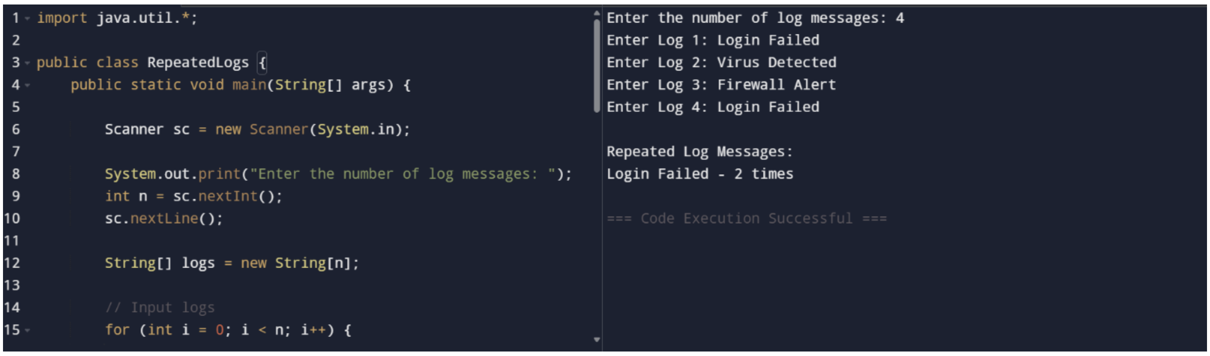
Implementation:



b) Navigate articles using ListIterator.

```
import java.util.*;

public class RepeatedLogs {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of log messages: ");
        int n = sc.nextInt();
        sc.nextLine();

        String[] logs = new String[n];

        // Input logs
        for (int i = 0; i < n; i++) {
            System.out.print("Enter Log " + (i + 1) + ": ");
            logs[i] = sc.nextLine();
        }

        // Find repeated logs
        System.out.println("\nRepeated Log Messages:");

        boolean found = false;

        for (int i = 0; i < n; i++) {
            int count = 1;

            if (logs[i].equals("Visited"))
                continue;
```

```
            for (int j = i + 1; j < n; j++) {
                if (logs[i].equals(logs[j])) {
                    count++;
                    logs[j] = "Visited";
                }
            }

            if (count > 1) {
                System.out.println(logs[i] + " - " + count + " times");
                found = true;
            }
        }

        if (!found) {
            System.out.println("No repeated log messages.");
        }

        sc.close();
    }
}
```



c) Compare String and StringBuilder

String	**StringBulider**
1) Immutable (cannot be changed).	1) Mutable (can be modified).
2) Creates a new object for every change..	2) Modifies the existing object.
3) Slower for repeated string operations.	3) Faster for repeated append and update operation
4) Uses concat() or + operator.	4) Uses `append()`, `insert()`, `delete()`, etc

c) **Explain Log Analysis**

Log analysis is the process of examining security event logs to identify suspicious activities, repeated events, errors, and security threats.

Importance of Log Analysis

- Detects unauthorized login attempts.
- Identifies repeated error or warning messages.
- Monitors system activities.
- Helps in troubleshooting system issues.
- Improves cyber security by detecting attacks early.

Example

Log Message	Analysis
Login Failed	Check for unauthorized access attempts.
Virus Detected	Scan and remove malware.
Firewall Alert	Investigate possible network attack.
Login Failed (Repeated)	May indicate a brute-force attack.